

DPP NO.-1

ELEMENTARY PROBLEMS ON PERMUTATION & COMBINATION

NOTE : USE FUNDAMENTAL PRINCIPLE OF COUNTING & ENJOY DOING THE FOLLOWING.

- Q.1 In how many ways can clean & clouded (overcast) days occur in a week assuming that an entire day is either clean or clouded.
- Q.2 Four visitors A, B, C & D arrive at a town which has 5 hotels. In how many ways can they disperse themselves among 5 hotels, if 4 hotels are used to accommodate them.
- Q.3 If the letters of the word “VARUN” are written in all possible ways and then are arranged as in a dictionary, then the rank of the word VARUN is :
(A) 98 (B) 99 (C) 100 (D) 101
- Q.4 How many natural numbers are there from 1 to 1000 which have none of their digits repeated.
- Q.5 A man has 3 jackets, 10 shirts, and 5 pairs of slacks. If an outfit consists of a jacket, a shirt, and a pair of slacks, how many different outfits can the man make?
- Q.6 There are 6 roads between A & B and 4 roads between B & C.
- (i) In how many ways can one drive from A to C by way of B?
- (ii) In how many ways can one drive from A to C and back to A, passing through B on both trips ?
- (iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.
- Q.7 (i) How many car number plates can be made if each plate contains 2 different letters of English alphabet, followed by 3 different digits.
(ii) Solve the problem, if the first digit cannot be 0. (Do not simplify)
- Q.8 (i) Find the number of four letter word that can be formed from the letters of the word HISTORY. (each letter to be used at most once)
(ii) How many of them contain only consonants?
(iii) How many of them begin & end in a consonant?
(iv) How many of them begin with a vowel?
(v) How many contain the letters Y?
(vi) How many begin with T & end in a vowel?
(vii) How many begin with T & also contain S?
(viii) How many contain both vowels?
- Q.9 If repetitions are not permitted
- (i) How many 3 digit numbers can be formed from the six digits 2, 3, 5, 6, 7 & 9 ?
- (ii) How many of these are less than 400 ?
- (iii) How many are even ?
- (iv) How many are odd ?
- (v) How many are multiples of 5 ?

- Q.10 In how many ways can 5 letters be mailed if there are 3 different mailboxes available if each letter can be mailed in any mailbox.
- Q.11 Every telephone number consists of 7 digits. How many telephone numbers are there which do not include any other digits but 2, 3, 5 & 7 ?
- Q.12 (a) In how many ways can four passengers be accommodated in three railway carriages, if each carriage can accommodate any number of passengers.
(b) In how many ways four persons can be accommodated in 3 different chairs if each person can occupy only one chair.
- Q.13 How many of the arrangements of the letter of the word "LOGARITHM" begin with a vowel and end with a consonant?
- Q.14 Number of natural numbers between 100 and 1000 such that at least one of their digits is 7, is
(A) 225 (B) 243 (C) 252 (D) none
- Q.15 How many four digit numbers are there which are divisible by 2 .
- Q.16 In a telephone system four different letter P, R, S, T and the four digits 3, 5, 7, 8 are used. Find the maximum number of "telephone numbers" the system can have if each consists of a letter followed by a four-digit number in which the digit may be repeated.
- Q.17 Find the number of 5 lettered palindromes which can be formed using the letters from the English alphabets.
- Q.18 Number of ways in which 7 different colours in a rainbow can be arranged if green is always in the middle.
- Q.19 Two cards are drawn one at a time & without replacement from a pack of 52 cards. Determine the number of ways in which the two cards can be drawn in a definite order.
- Q.20 Numbers of words which can be formed using all the letters of the word "AKSHI", if each word begins with vowel or terminates in vowel .
- Q.21 A letter lock consists of three rings each marked with 10 different letters. Find the number of ways in which it is possible to make an unsuccessful attempts to open the lock.
- Q.22 How many 10 digit numbers can be made with odd digits so that no two consecutive digits are same.
- Q.23 It is required to seat 5 men and 4 women in a row so that the women occupy the even places. How many such arrangements are possible?
- Q.24 If no two books are alike, in how many ways can 2 red, 3 green, and 4 blue books be arranged on a shelf so that all the books of the same colour are together?
- Q.25 How many natural numbers are there with the property that they can be expressed as the sum of the cubes of two natural numbers in two different ways.